

Direct equivalence of the two C4 rules

Figure1 and Figure1-MS, with scalars retained

Let p be an odd prime and $g \in \mathbb{F}_p^\times$ a primitive root. Write

$$M = M_g, \quad u = g^{-1}, \quad d = u^2, \quad b = g^2, \quad c = \frac{1-g}{2}d, \quad e = \frac{g^2-g}{2}.$$

Every displayed exponent of S or Z is an element of $\mathbb{F}_p$; the word S^a means the natural-number power at the representative of a in $\{0, \dots, p-1\}$. Fractions and inverses are computed in $\mathbb{F}_p$. Juxtaposition is word concatenation, with associativity and units suppressed. The symbol $\simeq$ denotes derivability from the specified axioms, and 1 denotes the empty word. These are exact equations, not equations modulo scalars.

Original C4 (Figure1) $MS \simeq Z^c S^d M$
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Replacement C4 (Figure1-MS) $SM \simeq MS^b Z^e$

1. Consequences of the shared axioms

Both presentations share C0–C3, C5–C15 and the structural circuit rules. The one-wire support needed below follows from C0–C3, C5 and scalar centrality:

$$S^p \simeq Z^p \simeq 1, \quad ZS \simeq SZ, \quad J^2 \simeq 1, \quad JM \simeq MJ, \quad J = H^2. \quad (1)$$

Here is the derivation of this support. C3 gives $M_g^k \simeq M_{g^k}$, $M_1 \simeq 1$ and $M_g^{p-1} \simeq 1$. Taking discrete logarithms of nonzero field elements and reducing exponents modulo $p-1$ gives

$$M_x M_y \simeq M_{xy}, \quad M_{-1}^2 \simeq 1, \quad M_{-1} M \simeq M M_{-1}.$$

Set $\lambda = (\nu^p)^{(p-1)/2}$, with ν the central scalar generator. C0 gives $\nu^{2p} \simeq 1$, hence $\lambda^2 \simeq 1$. C2 says $J \simeq M_{-1}\lambda$, so $J^2 \simeq 1$ and $JM \simeq MJ$.

Put $T = JSJ$. C5 says $ST \simeq TS$. Since $Z = JSJS^{-1} = TS^{-1}$, it follows that $ZS \simeq SZ$. Also $T^p \simeq JS^p J \simeq 1$ by C1 and $J^2 \simeq 1$; thus $Z^p \simeq T^p S^{-p} \simeq 1$. This proves (1) without either C4.

2. Deriving the action on Z from each C4

Define $\mathcal{C}(w) = JwJ$. By (1), this is an involutive word automorphism:

$$\mathcal{C}(xy) \simeq \mathcal{C}(x)\mathcal{C}(y), \quad \mathcal{C}^2(w) \simeq w, \quad \mathcal{C}(M) \simeq M.$$

The definition of Z gives $\mathcal{C}(S) \simeq ZS$. Applying $\mathcal{C}$ again yields $S \simeq \mathcal{C}(Z)ZS$. Cancelling the final S gives $\mathcal{C}(Z)Z \simeq 1$, so

$$\mathcal{C}(Z) \simeq Z^{-1}, \quad \mathcal{C}(Z^a S^h) \simeq Z^{h-a} S^h. \quad (2)$$

All cancellations here and below use the finite-order inverses from (1).

Starting with the original C4

Assume $MS \simeq Z^c S^d M$. Apply (2):

$$\begin{aligned} MZS &\simeq \mathcal{C}(MS) \simeq \mathcal{C}(Z^c S^d M) \simeq Z^{d-c} S^d M \\ &\simeq Z^{d-2c} (Z^c S^d M) \simeq Z^{d-2c} MS. \end{aligned}$$

Cancel S . Since

$$c = \frac{d-u}{2}, \quad d-2c = u,$$

we obtain $MZ \simeq Z^u M$. Iteration gives $MZ^x \simeq Z^{ux} M$. Taking $x = gk$ and using $ug = 1$ gives

$$\boxed{Z^k M \simeq MZ^{gk}}. \quad (3)$$

Starting with the replacement C4

Assume $SM \simeq MS^b Z^e \simeq MZ^e S^b$. Apply (2), then substitute the assumed rule:

$$\begin{aligned} ZSM &\simeq MZ^{b-e} S^b, \\ (ZM)(Z^e S^b) &\simeq ZSM \simeq MZ^{b-e} S^b \simeq (MZ^{b-2e})(Z^e S^b). \end{aligned}$$

Cancel $Z^e S^b$. As $b-2e = g$, this gives $ZM \simeq MZ^g$. Iteration again yields (3). In either theory, taking $k = ux$ in (3) also gives

$$\boxed{MZ^x \simeq Z^{ux} M}. \quad (4)$$

Thus each theory supplies (3)–(4) from its own C4; the argument is not circular.

3. Original C4 implies the replacement C4

Assume only the original C4 and the shared axioms:

$$MS \simeq Z^c S^d M.$$

Use (4), derived from this assumption on the preceding page.

Step 1: iterate the assumed rule $b = g^2$ times

Choose the natural representative of b for the induction. Because Z and S commute and have order p , their powers add and multiply in $\mathbb{F}_p$. Hence

$$\begin{aligned} MS^b &\simeq (Z^c S^d)^b M && \text{iterate the original C4} \\ &\simeq Z^{cb} S^{db} M && ZS \simeq SZ \text{ and power laws} \\ &\simeq Z^{cb} SM && db = u^2 g^2 = 1. \end{aligned}$$

Step 2: move the remaining Z -power through M

$$\begin{aligned} MS^b Z^e &\simeq Z^{cb} SM Z^e && \text{Step 1} \\ &\simeq Z^{cb} SZ^{eu} M && (4) \text{ with } x = e \\ &\simeq Z^{cb+eu} SM && ZS \simeq SZ \\ &\simeq SM && cb + eu = 0. \end{aligned}$$

The last field identity is

$$cb + eu = \frac{1-g}{2} u^2 g^2 + \frac{g^2-g}{2} u = \frac{1-g}{2} + \frac{g-1}{2} = 0.$$

Reverse the resulting equation to obtain exactly the replacement C4:

Derived in Figure1 $SM \simeq MS^{g^2} Z^{(g^2-g)/2}$

Formal counterpart

In `FF-Iso.agda`, the iteration and calculation above are `Conversion.MS  $\Rightarrow$  MR`. The theorem `Figure1.c4'` reverses its conclusion to match the orientation of the Figure1-MS axiom.

4. Replacement C4 implies the original C4

Assume only the replacement C4 and the shared axioms:

$$SM \simeq MS^b Z^e.$$

Use (4), now derived from this assumption. Set $f = ed$.

Step 1: iterate the assumed rule $d = g^{-2}$ times

Again choose the natural representative for the induction. Then

$$\begin{aligned} S^d M &\simeq M(S^b Z^e)^d && \text{iterate the replacement C4} \\ &\simeq MS^{bd} Z^{ed} && ZS \simeq SZ \text{ and power laws} \\ &\simeq MSZ^f && bd = g^2 u^2 = 1. \end{aligned}$$

Step 2: remove the extra Z -power

$$\begin{aligned} MS &\simeq (MSZ^f)Z^{-f} && \text{insert } Z^f Z^{-f} \simeq 1 \\ &\simeq S^d MZ^{-f} && \text{Step 1} \\ &\simeq S^d Z^{-fu} M && (4) \text{ with } x = -f \\ &\simeq Z^c S^d M && ZS \simeq SZ \text{ and } -fu = c. \end{aligned}$$

The last field identity is

$$-fu = -edu = -\frac{g^2 - g}{2} u^3 = \frac{u^2 - u}{2} = \frac{1 - g}{2} u^2 = c.$$

This is exactly the original C4:

Derived in Figure1-MS $MS \simeq Z^{(1-g)g^{-2}/2} S^{g^{-2}} M$

Formal counterpart and equivalence

This direction is `Conversion.MR  $\Rightarrow$  MS` and `Figure1-MS.c4` in `FF-Iso.agda`. Every other axiom is carried over unchanged, and derivations are transported recursively through the congruence and structural rules. Therefore the two presentations generate the same equivalence relation on the same circuits. Their presented groups are isomorphic by the identity on words.

Formal source: `Examples/Groups/Clifford+MinusOne/Qupit/Figure1-MS/FF-Iso.agda`. The Agda file proves both directions syntactically, without importing Clifford semantics or the existing exact-group isomorphisms.